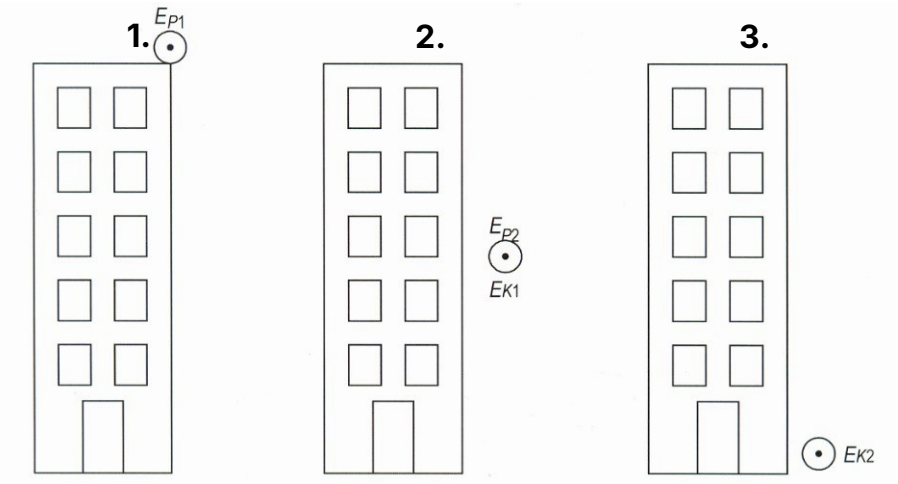
## Calculating energy transfers

It has already been stated that energy cannot be created or destroyed – it can only be transformed to another energy. During an energy transformation, the total energy contained within any closed system remains constant. Knowing the total amount of energy at the start (or end) of any energy transformation tells us the total energy at any given time during the transformation.

### Example

**1.** Consider a ball that is released from a high building. When the ball is at the top of the building, it has gravitational potential energy ($E_{P1}$)




**The total energy, $E_T$ = $E_{P1}$**

**2.** When the ball is released, it falls and some of the potential energy ($E_{P2}$) is converted into kinetic energy ($E_{K1}$). The total amount of energy ($E_T$) the ball possesses at this time is equal to the potential energy plus the kinetic energy.

**The total energy $E_T = (E_{P2} + E_{K1})$**

**3.** Just as the ball hits the ground, it no longer has any potential energy – all the potential energy ($E_{P1}$) has been converted into kinetic energy ($E_{K1}$). The total amount of energy now consists of the kinetic energy alone.

**Total energy $E_T = E_{P1} = (E_{P2} + E_{K1}) = E_{k2}$**

## Worked example: transfer of energy (potential to kinetic)

A body of mass 30kg falls freely from a height of 20 metres. Find its final velocity and kinetic energy at impact. First calculate the initial potential energy.

$E_P = mgh$

$= 30 \times 9.8 \times 20$

$= 5880J \quad = 5.9\ kJ$

This potential energy is converted or transferred into kinetic energy, which means that the kinetic energy at impact is equal to 5.9 kJ. But this is not the end of the calculation!

To calculate the final velocity of the body we begin by taking $E_k$ = 5.88 kJ.

$E_K = \frac{1}{2}mv^2$

$5.88 \times 10^3 = \frac{1}{2} \times 30 \times v^2$

$5880 = 15 \times v^2$

$v^2 = 5880 / 15$

$v^2 = 392$

$v = \ = 19.79\ m/s$

This type of calculation can be completed for any type of energy conversion. Knowing the total amount of energy at any given time (beginning/middle/end) is the same at any other given time.

**Q1.** At the top of a 50cm ramp a 1.5kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill. Calculate the transfer of energy.

**Q2.** At the top of a 80cm ramp, a 3kg buggy rolls down to the bottom using gravity. Work out the potential and kinetic energy at the top and middle of the hill.

**Q3.** A 5kg mass is raised steadily through a height of 2m. What work is done and what is the body's potential energy relative to the start?

**Q4.** A mass of 20kg is allowed to fall freely from a certain height above a datum. When the body is 16m above the datum, it possesses a total energy of 3531J. What is the starting height of the object?